

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Mixed

10 answers · Rationale-rich · Teach from doc

04

Q1**D**

D. No additional information is needed.

Choice D is correct. When two angles of one triangle are congruent to two angles of another triangle, the triangles are similar. It's given that in triangle DEF , the measure of angle D is 47° and the measure of angle E is 97° . It's also given that in triangle RST , the measure of angle R is 47° and the measure of angle S is 97° . It follows that angle D is congruent to angle R and that angle E is congruent to angle S . Therefore, triangle DEF is similar to triangle RST and no additional information is needed.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q2**21**

The correct answer is 21. It's given that triangles ABC and DEF are congruent with angle A corresponding to angle D . Corresponding angles of congruent triangles are congruent and, therefore, have equal measure. It's given that the measure of angle A is 69° . It follows that the measure of angle D is also 69° . It's given that angle E is a right angle. Therefore, the measure of angle E is 90° . Let x represent the measure, in degrees, of angle F . Since the measures of the angles in a triangle sum to 180° , it follows that $69 + 90 + x = 180$, or $159 + x = 180$. Subtracting 159 from both sides of this equation yields $x = 21$. Therefore, the measure, in degrees, of angle F is 21.

Q3**C**

C. Infinitely many

Choice C is correct. If the two sides of a linear equation are equivalent, then the equation is true for any value. If an equation is true for any value, it has infinitely many solutions. Since the two sides of the given linear equation $66x = 66x$ are equivalent, the given equation has infinitely many solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4**A**

A. 2

Choice A is correct. Adding the like terms on the left-hand side of the given equation yields $5(x - 5) = 10$. Dividing both sides of this equation by 5 yields $x - 5 = 2$.

Choice B is incorrect and may result from subtracting 5, not dividing by 5, on both sides of the equation $5(x - 5) = 10$. Choice C is incorrect. This is the value of x , not the value of $x - 5$.

Choice D is incorrect. This is the value of $x + 5$, not the value of $x - 5$.

Q5**39**

The correct answer is 39. It's given that for the linear function j , m is a constant and $j12 = 18$. Substituting 12 for x and 18 for jx in the given equation yields $18 = m12 + 144$. Subtracting 144 from both sides of this equation yields $-126 = m12$. Dividing both sides of this equation by 12 yields $-10.5 = m$. Substituting -10.5 for m in the given equation, $jx = mx + 144$, yields $jx = -10.5x + 144$. Substituting 10 for x in this equation yields $j10 = -10.5(10) + 144$, or $j10 = 39$. Therefore, the value of $j10$ is 39.

Q6**D**

D. $\frac{18x^2}{5} + 4x - \frac{35}{6}$

Choice D is correct. It's given that $gx = \frac{3}{5}x + \frac{7}{6}$ and $hx = 6x - 5$. Substituting $\frac{3}{5}x + \frac{7}{6}$ for gx and $6x - 5$ for hx in the expression $gx \cdot hx$ yields $\frac{3}{5}x + \frac{7}{6}6x - 5$. This expression can be rewritten as $\frac{3}{5}x6x - 5 + \frac{7}{6}6x - 5$, or $\frac{18x^2}{5} - 3x + 7x - \frac{35}{6}$, which is equivalent to $\frac{18x^2}{5} + 4x - \frac{35}{6}$.

Choice A is incorrect. This expression is equivalent to $\frac{3}{5}x6x + \frac{7}{6}5$, not $\frac{3}{5}x + \frac{7}{6}6x - 5$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This expression is equivalent to $\frac{3}{5}x - \frac{7}{6}6x + 5$, not $\frac{3}{5}x + \frac{7}{6}6x - 5$.

Q7**D**

D. The estimated initial population for this species and area in 2008

Choice D is correct. Substituting 0 for x in the given equation yields $f0 = 3,0000.75^0$, which is equivalent to $f0 = 3,0001$, or $f0 = 3,000$. It's given that the function estimates the species' population x years after 2008, so it follows that the estimated population of the species is 3,000 in 2008. Therefore, the best interpretation of 3,000 in this context is the estimated initial population for this species and area in 2008.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. The estimated percent decrease in the population for this species and area each year after 2008 is 25%, not 3,000.

Choice C is incorrect. The estimated population for this species and area 8 years after 2008 is $3,0000.75^8$, or approximately 300, not 3,000.

Q8**3**

The correct answer is 3. The solution to the given equation can be found by factoring the quadratic expression. The factors can be determined by finding two numbers with a sum of 1 and a product of -12 . The two numbers that meet these constraints are 4 and -3 . Therefore, the given equation can be rewritten as $(x+4)(x-3)=0$. It follows that the solutions to the equation are $x=-4$ or $x=3$. Since it is given that $a>0$, a must equal 3.

Q9**B**

B. Between 28.59% and 35.41% of all adults who own televisions use their televisions to watch nature shows.

Choice B is correct. It's given that based on a sample selected at random, it's estimated that 32% of all adults who own televisions use their televisions to watch nature shows, with an associated margin of error of 3.41%. Subtracting the margin of error from the estimate and adding the margin of error to the estimate gives an interval of plausible values for the true percentage of adults who own televisions who use their televisions to watch nature shows. This means it's plausible that between $32\% - 3.41\%$, or 28.59%, and $32\% + 3.41\%$, or 35.41%, of all adults who own televisions use their televisions to watch nature shows. Therefore, of the given choices, the most plausible conclusion is that between 28.59% and 35.41% of all adults who own televisions use their televisions to watch nature shows.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. To make a plausible conclusion about all adults who own televisions, the sample must be selected at random from all adults who own televisions, not just those who use their televisions to watch nature shows.

Choice D is incorrect. Since the sample was selected at random from all adults who own televisions, a plausible conclusion can be made about all adults who own televisions.

Q10

27

The correct answer is 27. It's given that 6% of the 450 tiles in a box are black. Therefore, the number of black tiles in the box can be calculated by multiplying the number of tiles in the box by $\frac{6}{100}$, which is equivalent to $450\frac{6}{100}$, or 27.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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